Time Domain Specifications, Dominant Mode and Additional Poles and Zeros

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1 Introduction

In this lecture we will consider:

1. Standard controller specifications
2. Derivation of Peak Time, Percent Overshoot, Rise Time and Settling Time
3. Converting specifications to allowed regions for poles on the S-plane
4. The effects of additional zeros and poles
   (a) Dimensional analysis
   (b) Effects of a zero
   (c) Effects of an additional pole
   (d) Summary, Effects of a zero or an additional pole
5. Stability and the Routh stability criterion

This lecture relates to Franklin *et al.* sections 3.4 - 3.6 (7th ed.).

1.1 Design cycle, with approximate and exact calculations

The process typical of control system design is illustrated in figure 1. The design cycle comprises two parts:

1. A design phases linking performance goals to the design choices. For this there are generally no exact calculations, so approximation connections between engineered characteristics, such as closed-loop pole locations and rise time or overshoot, are used.

2. An analysis phase, to verify that the completed design meets the specifications. Exact calculations, such as calculation of the step response to get rise time and overshoot, are used.

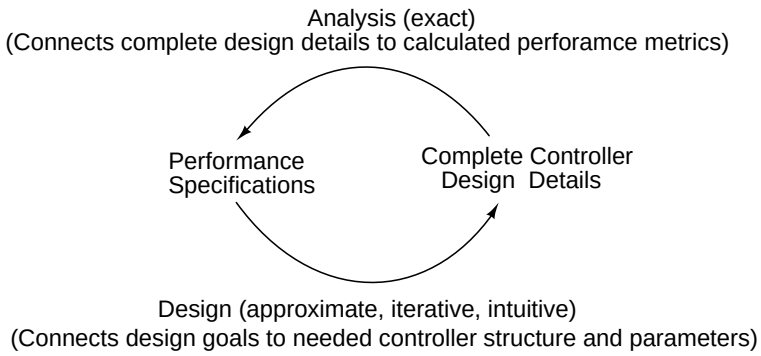


Figure 1: A controller design cycle combines a design phase with an analysis phase.

- In the coming sections we will develop approximate performance calculations, that can guide the design phase.

2 Controller Specifications

- Controller specifications are quantitative measures of closed-loop dynamic system performance.
  - Controller specs are used to describe or specify controller performance.
  - We need a way to compute each of these, based on the model of the system and description of the controller.
- Controller specifications commonly used in industry are seen in table 1. Big view
  - We will work with the eight underlined specifications in EE 474.
  - We will work with all fifteen specifications in EE 574.

	Speed	Degree of Stability	Accuracy
Time Domain	* Rise Time * Settling Time * Peak Time	* Overshoot * Damping Factor	* Steady-State Error
Frequency Domain (Or S-plane)	* Pole Locations * Bandwidth * Cross-over Freq.	* Pole Locations * Phase Margin * Gain Margin	* Disturbance Rejection * Noise Rejection * Dynamic Tracking Acc.

Table 1: A range of controller specifications used in industry.
Note: Underlined items developed and applied in EE 474.

2.1 Basic analysis tool of controller specs

- We've seen that there are various ways to represent a control system: Big view

  1. Complete transfer function;
  2. Pole locations of dominant mode;
  3. Step or impulse response (considering dominant mode);
  4. Complete pole-zero constellation.
  5. Open-loop frequency response / Bode plot.
  6. Closed-loop frequency response.

- As seen in table 2, each of the quantitative specifications is based on a specific modeling representation.
  - Several can be based on a dominant mode approximation (3), or exactly computed from the complete transfer function (1).

- All model representations needed:
  - Five of six are used to compute at least one specs.
  - The 6th is needed for the Root Locus method.
  - Thus the need to convert between modeling representations.

	Speed	Degree of Stability	Accuracy
Time Domain	* Rise Time (1,3) * Settling Time (1,3) * Peak Time (1,3)	* Overshoot (1,3) * Damping Factor(2)	* Steady-State Error (1)
Frequency Domain (Or S-plane)	* Pole Locations (2) * Bandwidth (5) * Cross-over Freq.(5)	* Pole Locations (2) * Phase Margin (5) * Gain Margin (5)	* Disturbance Rejection (6) * Noise Rejection (6) * Dynamic Tracking Acc. (5)

Table 2: Analysis tool for each specification.

2.2 Controller specifications derived in this lecture

- In this lecture, we will derive analysis equations for specifications based on the step response of the dominant mode, model representation (3).
  - We will consider the response of the dominant mode (an approximation).
  - As shown in table 2, these properties can also be determined exactly for the complete TF, (1).
- Why use the step response of the dominant mode (3), an approximation ? Big view
  - Approximate, but gives insight into controller design.
  - The controller design problem is often:
    - Not a question of setting a parameter at 5 or 6;
    - But a question of setting a parameter at 0.05, 5 or 500;
    - Or of controller structure in the first place: such as P, PI, PID or other.
  - The dominant mode approximation is often as close as practical.
    - Recall that we rarely know system parameters exactly (how much does the car weigh when the cruise control is used, anyway).
- When use the complete TF (1, e.g., `step` command in Matlab) ?
  - For an exact numerical solution;
  - When insight is not needed;
  - To fine tune, when controller structure and coarse parameter values have been determined.
  - To validate that the completed system will meet specification.

	Speed	Degree of Stability	Accuracy
Time Domain	* **Rise Time** * **Settling Time** * **Peak Time**	* **Percent Overshoot** * Damping Factor	* *Steady-State Error*
S-plane	* *Pole Locations*	* *Pole Locations*	

Table 3: Controller specifications: **bold** items are developed here; *italics* items have been developed in previous lectures.

3 Derivation of Peak time and Percent Overshoot, Rise time and Settling time from Dominant Mode

- As described in the previous lecture, the dominant mode will either be first or second order.
  - For reasons we'll see soon, many closed-loop systems have a dominant second-order mode.
  - The second order is considered first, and then first order characteristics appear as a special case.

3.1 Characteristics of the second order dominant mode step response

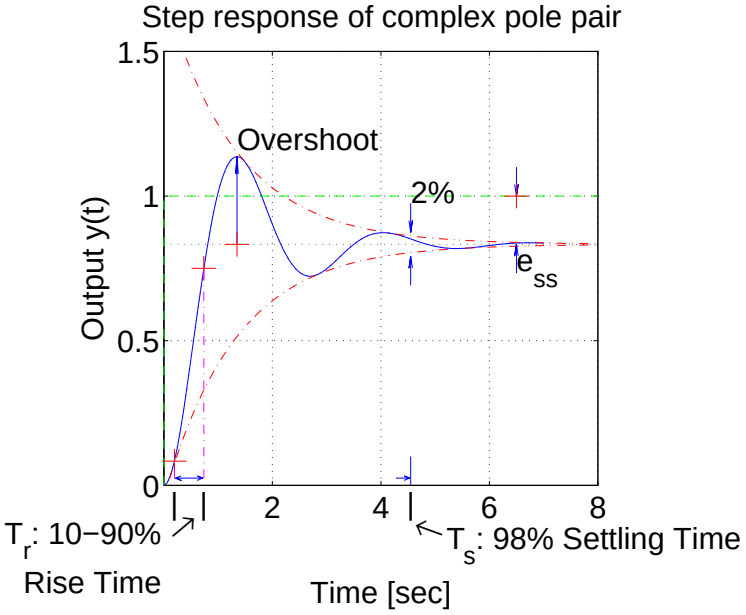


Figure 2: The step response of the example system.

- These characteristics are illustrated in figure 2:
  - T_r: the 10%-90% rise time, $T_r = 0.54\,[\text{s}]$;
  - T_s: the settling time, $T_s = 4.49\,[\text{s}]$;
  - Percent Overshoot: $PO = 36.4\%$.
  - e_{ss}: the steady state error, $e_{ss} = 16.7\%$.

3.1.1 Computing various magnitudes and times in a step response

- A step response generates a somewhat elaborate set of measurements. Figure 4 shows ten measurements and one computed value.

  These are listed in table 4.

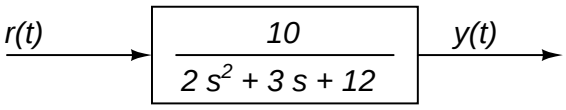


Figure 3: The step response of an example system with offsets, example system.

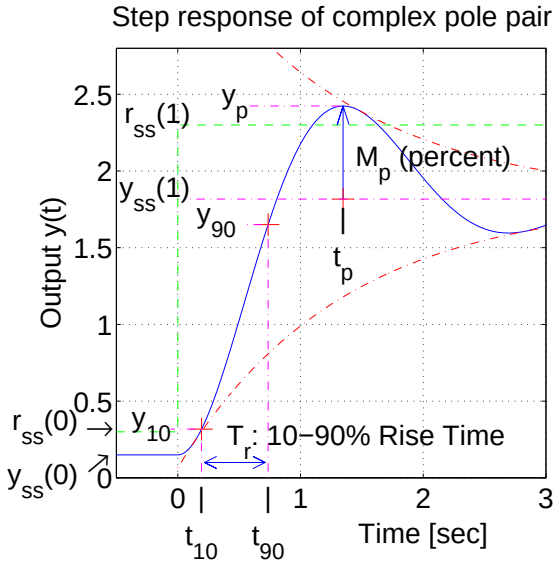


Figure 4: The step response of an example system with offsets.

Measurements				
Reference Signal Steady State	Output Signal Steady State	Peak	10% Trans.	90% Trans.
before: $r_{ss}(0)$	before: $y_{ss}(0)$	t_p	t_{90}	t_{10}
after: $r_{ss}(1)$	after: $y_{ss}(1)$	y_p	y_{90}	y_{10}
		M_p		

Table 4: List of measurements illustrated in figure 3.

Computing various magnitudes and times in a step response (cont.)

- Figure 4 is the general case, more general than is often shown, but typical of actual data.
  - The Initial values are not zero.
  - The step size is not one.
- Steady States:

  We have steady states before and after the transient, and of the reference and output signals. The changes in steady-state signals are given by:

$$\Delta_r = r_{ss}(1) - r_{ss}(0) = 2.0 \quad ; \qquad \Delta_y = y_{ss}(1) - y_{ss}(0) = 5/3 \tag{1}$$

- The peak:

  The peak time and value are given as t_p and y_p respectively. The peak magnitude is expressed as a percentage change:

$$M_p = 100 \times \frac{y_p - y_{ss}(1)}{\Delta_y} = 100 \times \frac{y_p - y_{ss}(1)}{y_{ss}(1) - y_{ss}(0)} \qquad \left[\text{dim}'\text{less}\right] \tag{2}$$

- 10% and 90% levels:

  The 10% and 90% levels are relative to the initial steady state, $y_{ss}(0)$, and the size of the transient:

$$\begin{aligned} y_{10} &= y_{ss}(0) + 0.10\Delta_y \\ y_{90} &= y_{ss}(0) + 0.90\Delta_y = y_{ss}(1) - 0.10\Delta_y \end{aligned} \tag{3}$$

  Times t_{10} and t_{90} can be measured from experimental data or determined from a computed response, from $t_{10}: \; y(t_{10}) = y_{10}$ and $t_{90}: \; y(t_{90}) = y_{90}$.

- DC gain:

  The DC gain of the system is directly observed in the figure:

$$K_{DC} = \frac{\Delta_y}{\Delta_r} \qquad [\text{units of gain}] \tag{4}$$

3.2 Quantitative specifications from dominant mode

- **Goal:** Derive equations to link pole locations to quantitative specifications.
  - Eqns should be reasonably simple, to be useful during controller design
  - Complete pole-zero constellation is too complex to derive reasonably simple expressions
- **Simplifying assumption:** dominant mode is approximated by a pole pair with a DC gain of K_{DC} and no zeros:

$$H(s) = K_{DC} \frac{\omega_n^2}{s^2 + 2\alpha s + \omega_n^2} \tag{5}$$

where

$$K_{rlg} = K_{DC}\, \omega_n^2 \tag{6}$$

- **First step:** find an expression for t_p: the peak time.

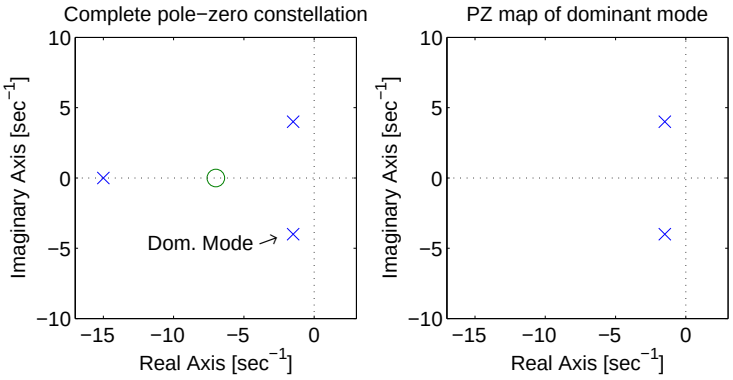


Figure 5: Example case, complete pole-zero constellation and dominant mode.

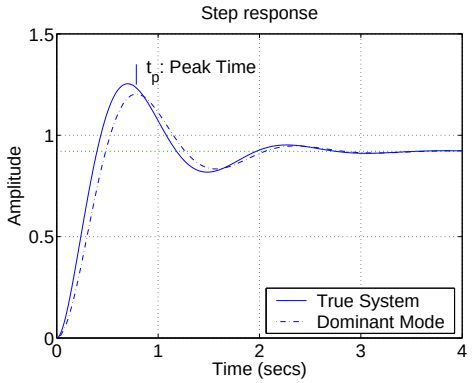


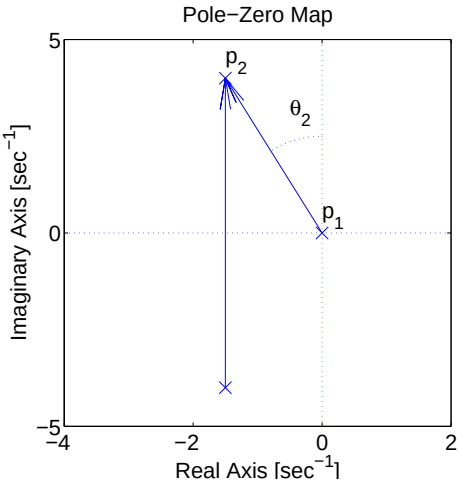
Figure 6: Example case, step response of p-z constellation and dominant mode. **Note:** t_p and M_p are determined with respect to dominant mode.

3.2.1 Determining peak time, t_p (Franklin *et al.* section 3.4 (7th ed.))

- Considering the step response combines:
  - The complex poles of the dominant mode; and
  - The pole at the origin from the step input:

$$Y(s) = H(s)\ U(s) = K_{DC}\ \frac{\omega_n^2}{s^2 + 2\alpha s + \omega_n^2}\ \frac{1}{s} \tag{7}$$

and gives the pole-zero plot of figure 7.



Notes
Poles at:

$$p_{1,2} = -\alpha \pm j\,\omega$$

Root locus gain:

$$K_{rlg} = K_{DC}\ \omega_n^2$$

Figure 7: Pole-Zero map for determining 2nd order dominant mode response.

- The oscillation of the response is contributed by the complex pole pair, with residue given by the two graphical contributions:

$$\begin{aligned} C_2 &= K_{rlg}\ \frac{1}{(\omega_n \angle(\theta_2 + 90^o)) \times (2\,\omega \angle 90^o)} \\ &= K_{DC}\ \omega_n^2 \left(\frac{1}{2\,\omega_n\,\omega}\right) \angle -(\theta_2 + 180^o) = K_{DC}\ \frac{1}{2}\ \frac{\omega_n}{\omega} \angle \phi_2 \end{aligned} \tag{8}$$

The angle ϕ_2 will be a large negative angle, so we can add 360°:

$$\phi_2 = -\theta_2 - 180^o + 360^o = 180^o - \theta_2 \tag{9}$$

Determining peak time (continued)

- Using the trig identities:

$$\sin(\phi_2) = \sin(180^o - \theta_2) = \sin(180^o)\cos(\theta_2) - \cos(180^o)\sin(\theta_2) \quad (10)$$
$$\cos(\phi_2) = \cos(180^o - \theta_2) = \cos(180^o)\cos(\theta_2) + \sin(180^o)\sin(\theta_2) \quad (11)$$

- One finds:

$$\begin{aligned} \sin(\phi_2) &= \sin(\theta_2) = \frac{\alpha}{\omega_n} \\ \cos(\phi_2) &= -\cos(\theta_2) = -\frac{\omega}{\omega_n} \end{aligned} \quad (12)$$

  where the last identities are seen in the geometry of figure 7.

- And so residue C_2 represented in rectangular coordinates is:

$$\begin{aligned} C_2 &= \frac{1}{2} K_{DC} \frac{\omega_n}{\omega} \angle \phi_2 \\ &= c_r + j c_j \end{aligned} \qquad \begin{aligned} c_r &= \frac{1}{2} K_{DC} \frac{\omega_n}{\omega} \cos(\phi_2) = -\frac{1}{2} K_{DC} \\ c_j &= \frac{1}{2} K_{DC} \frac{\omega_n}{\omega} \sin(\phi_2) = \frac{1}{2} K_{DC} \frac{\alpha}{\omega} \end{aligned} \quad (13)$$

- Considering that we can write:

$$C_2 = c_r + j c_j \iff y_2(t) = 2 c_r \cos(\omega t) - 2 c_j \sin(\omega t)$$

  The time domain expression for the decaying exponential part is:

$$y_2(t) = -K_{DC}\, e^{-\alpha t} \left(\cos(\omega t) + \frac{\alpha}{\omega} \sin(\omega t) \right) \quad (14)$$

Determining peak time (continued)

- The pole at the origin must also be considered. With Eqn (6) one finds:

$$C_1 = K_{DC}\,\omega_n^2\,\frac{1}{\omega_n\angle-(\theta_2+90^o)\;\omega_n\angle+(\theta_2+90^o)} = K_{DC}\,\angle 0^o \qquad (15)$$

  with the corresponding expression of the steady-state output:

$$y_1(t) = K_{DC}\,1^+(t) = y_{ss} \qquad (16)$$

  This is logical, since K_{DC} gives the magnitude of the steady-state output for a unit step input.

- Putting the pieces together $y(t)$ is give by:

$$y(t) \;=\; K_{DC}\,1^+(t) \;-\; K_{DC}\,e^{-\alpha t}\left(\cos(\omega t)+\frac{\alpha}{\omega}\sin(\omega t)\right) \qquad (17)$$

Now solve for t_p

- To find t_p, take the derivative of $y(t)$:

$$\begin{aligned}\dot{y}(t) = 0 \;+\; & K_{DC}\,\alpha\,e^{-\alpha t}\left(\cos(\omega t)+\frac{\alpha}{\omega}\sin(\omega t)\right)\\ & -\quad K_{DC}\,e^{-\alpha t}\left(-\omega\sin(\omega t)+\alpha\cos(\omega t)\right)\end{aligned}$$

  The cosine terms drop out, so:

$$\dot{y}(t) \;=\; K_{DC}\left(\frac{\alpha^2}{\omega}\sin(\omega t)+\omega\sin(\omega t)\right) \qquad (18)$$

Determining peak time (continued)

- The peak of the transient occurs when $\dot{y}(t) = 0$, which occurs when $\sin(\omega t) = 0$. Since $\sin(\pi) = 0$, the peak time is given by:

$$\omega\, t_p = \pi \qquad \text{or} \qquad t_p = \frac{\pi}{\omega} \tag{19}$$

!!!! After all that algebra, this is a remarkably simple result. !!!!

- In the example of page 11, which had a complex pole pair with $\omega = 4\,[\text{rad/sec}]$, the peak overshoot occurs at:

$$t_p = \frac{\pi\ [\text{rad}]}{4\ [\text{rad/sec}]} = 0.785\ [\text{sec}]$$

which is confirmed in figure 6.

3.2.2 Using t_p to determine the overshoot and interpret damping factor ρ

- Substituting $t_p = \pi/\omega$ into Eqn (17) for $y(t)$ gives:

$$\begin{aligned} y(t_p) &= K_{DC}\left(1^+(t_p) - e^{-\alpha\frac{\pi}{\omega}}\left(\cos\left(\omega\frac{\pi}{\omega}\right)+\frac{\alpha}{\omega}\sin\left(\omega\frac{\pi}{\omega}\right)\right)\right) \quad (20)\\ &= K_{DC}\left(1^+(t_p) - e^{-\alpha\frac{\pi}{\omega}}\left(\cos(\pi)+\frac{\alpha}{\omega}\sin(\pi)\right)\right)\\ &= K_{DC}\left(1^+(t_p) + e^{-\alpha\frac{\pi}{\omega}}\right) \quad (21)\end{aligned}$$

- M_p , the Magnitude of the Peak, gives the overshoot and is given in percent as:

$$M_p = 100\% \times \frac{y_p - y_{ss}}{y_{ss}}$$

  Notice that M_p is dimensionless.

- Plugging expressions (16) and (21), gives:

$$M_p = 100\% \times \frac{K_{DC}\left(1^+(t_p) + e^{-\alpha\frac{\pi}{\omega}}\right) - K_{DC}}{K_{DC}} = 100\% \times e^{-\pi\frac{\alpha}{\omega}} \quad (22)$$

- Now using the relationship between a pole expressed in rectangular $\left\{\ \alpha,\ \ \omega\ \right\}$ and polar $\left\{\ \rho,\ \ \omega_n\ \right\}$ coordinates

$$\rho = \frac{\alpha}{\omega_n}, \qquad \omega = \sqrt{1-\rho^2}\,\omega_n \quad (23)$$

  gives

$$M_p = 100\% \times e^{-\pi\frac{\alpha}{\omega}} = 100\% \times e^{-\frac{\pi\rho}{\sqrt{1-\rho^2}}} \quad (24)$$

- With the dominant mode approximation, the magnitude of the overshoot depends only on the damping factor, ρ !

Using t_p to determine the overshoot and interpret ρ(continued)

- Eqn (24) is evaluated to make figure 8, with overshoot expressed as percent.

$$M_p = 100\% \times e^{-\frac{\pi\rho}{\sqrt{1-\rho^2}}} \qquad \text{(24, repeated)}$$

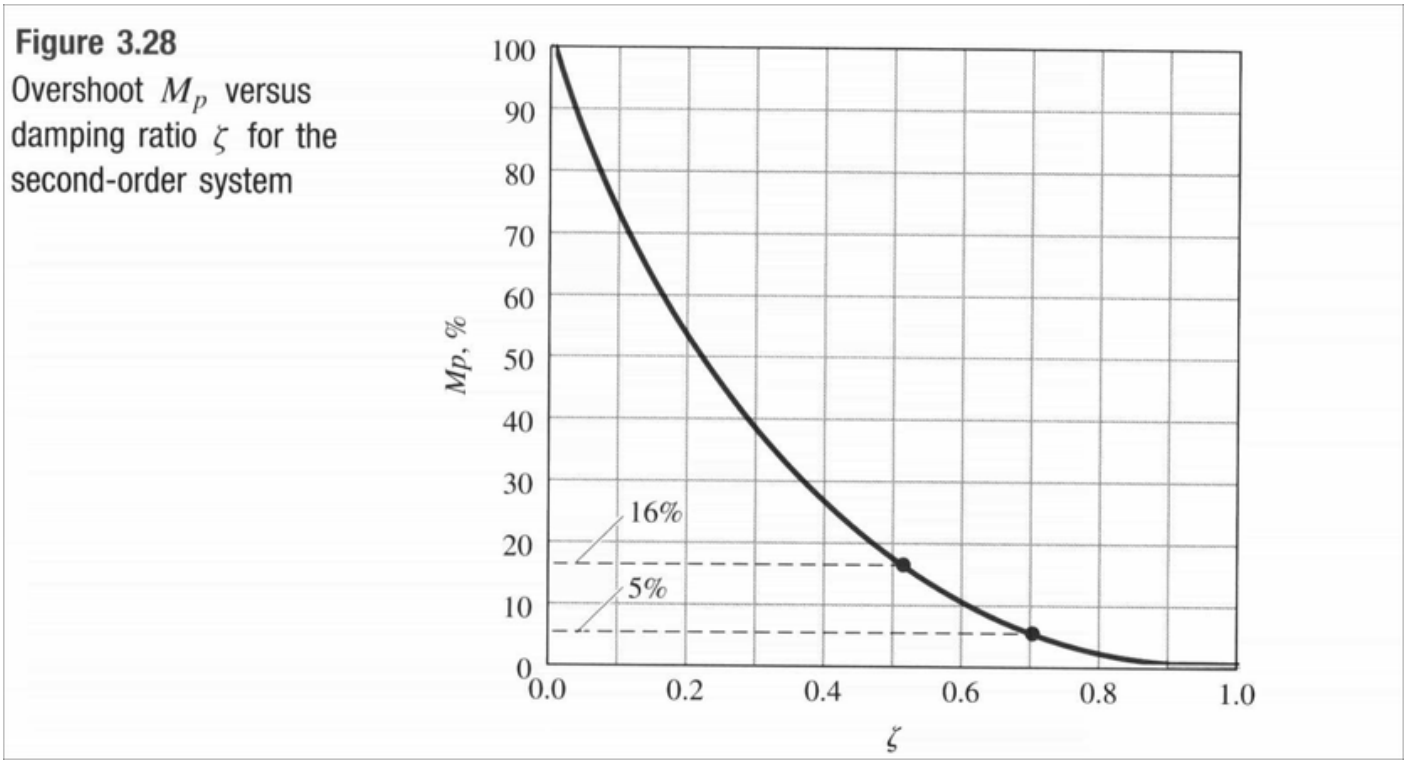


Figure 8: Chart of overshoot versus damping (Exact for a complex pole pair with no zero, otherwise approximate).

- It is also interesting to be able to determine the needed damping factor from a specified maximum percent overshoot. Eqn (22) can be rearranged to give:

$$\rho = \sqrt{\frac{\ln\left(M_p/100\right)^2}{\pi^2 + \ln\left(M_p/100\right)^2}} \qquad (25)$$

3.2.3 Estimating the rise time, t_r

- The 10-90 rise time, as is seen in figure 2, is the time required to transit from 10% of the full transit, to 90% of the full transit.

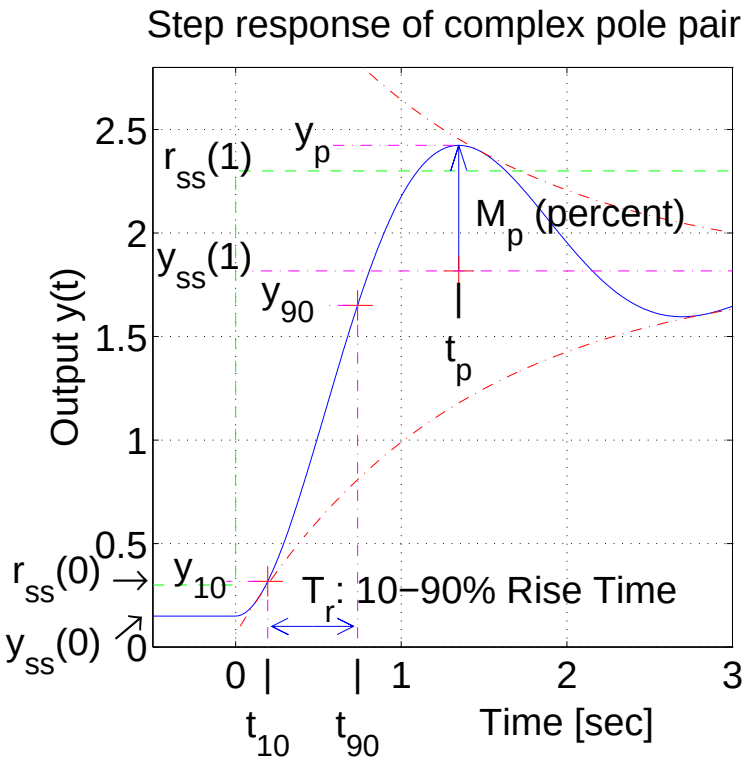
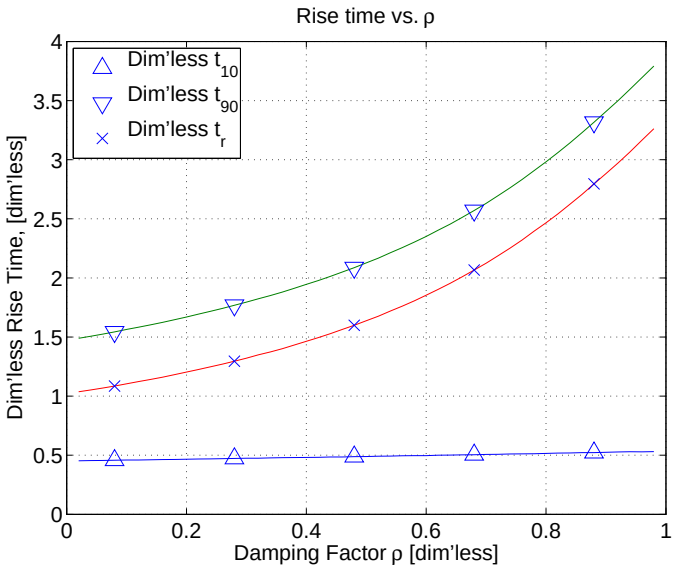


Figure 9: The step response of the example system.

- The rise time, t_r, is a measure of how fast the system gets near the goal, it is important, for example, for some high-speed tracking systems (e.g., Lab part 7)
  - We know that time scales with the radial distance of poles on the S-plane, so we can normalize t_r with respect to radial distance of the poles:

$$t_r^* = t_r\,\omega_n \qquad [\text{sec}]\,[\text{rad/sec}] = [\text{dim'less}] \tag{26}$$

  - The normalized rise time, t_r^*, will depend only the damping, and is seen in figure 10.



$$t_r = \frac{t_r^*}{\omega_n}\,[sec]$$

$$t_r^* = t_r\,\omega_n\,[\cdot]$$

Figure 10: Normalized rise time versus damping factor.

3.2.3.1 An approximation for rise time

- The closed-form solution for the curves of figure 10 is too complex to give insight.
  - Figure 10 shows the rise time can vary from $1.0/\omega_n$ to $3.0/\omega_n$, and that:
    - t_r goes up as damping increases.
  - Eqn (26) relates t_r to ω_n, which varies over orders of magnitude.
- A commonly used approximation for t_r, that captures the important connection to ω_n is:

$$t_r^* \simeq 1.8 \qquad \text{giving} \qquad t_r \simeq \frac{1.8}{\omega_n}\ \text{[sec]} \tag{27}$$

- Interesting note: for fast response, the controls of fighter jets are often tuned to be lightly damped.
  - The airframe of the F-16 was designed to be open-loop unstable, and the aircraft is stabilized by the flight controls.

    This way, the inherent stability of the airframe doesn't get in the way of very rapid response.

3.2.4 Estimating the settling time, t_s

- Figure 2 is repeated here.

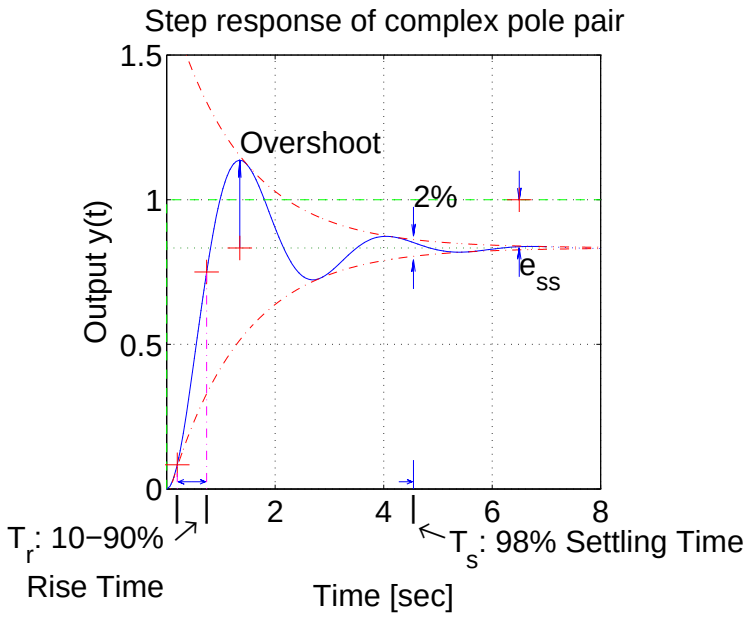


Figure 11: The step response of the example system.

- The dashed line forms an envelope in which the transient decays. The transient decays as $e^{-\alpha t}$.
  - For decay to 2% of the initial transient this gives:

$$e^{-\alpha t_s} = 0.02 \simeq e^{-4}$$

    which gives:

$$t_s \simeq \frac{4}{\alpha} = \frac{4}{\rho\,\omega_n} \quad [\text{sec}] \tag{28}$$

  - For decay to 1% of the initial transient (2% is more often used) the decay envelop gives:

$$e^{-\alpha t_s} = 0.01 \simeq e^{-4.6}$$

    which gives:

$$t_s \simeq \frac{4.6}{\alpha} = \frac{4.6}{\rho\,\omega_n} \quad [\text{sec}] \tag{29}$$

3.3 Summary and First Order Dominant Mode

- Putting together the information from the above sections for 2nd order systems, and completing the table for 1st order systems gives table 5.

	Order of Dominant Mode	
Parameter	1st Order	2nd Order
t_p	-	$t_p = \frac{\pi}{\omega}$
M_p (or P.O.)	-	$M_p = 100\% \times e^{-\frac{\pi\rho}{\sqrt{1-\rho^2}}}$
t_r	$t_r \simeq \frac{2.2}{\alpha}$	$t_r \simeq \frac{1.8}{\omega_n}$
t_s	$t_s \simeq \frac{4}{\alpha}$	$t_s \simeq \frac{4}{\alpha}$

Table 5: Summary of quantitative performance specifications.

4 Converting specifications to Regions on the S-plane

- Requirements on overshoot or damping, rise time, and settling time create allowed regions on the S-plane.

Constant:	ω_n	ρ	α	
Factor Set:	t_r	P.O.	t_s	All 3

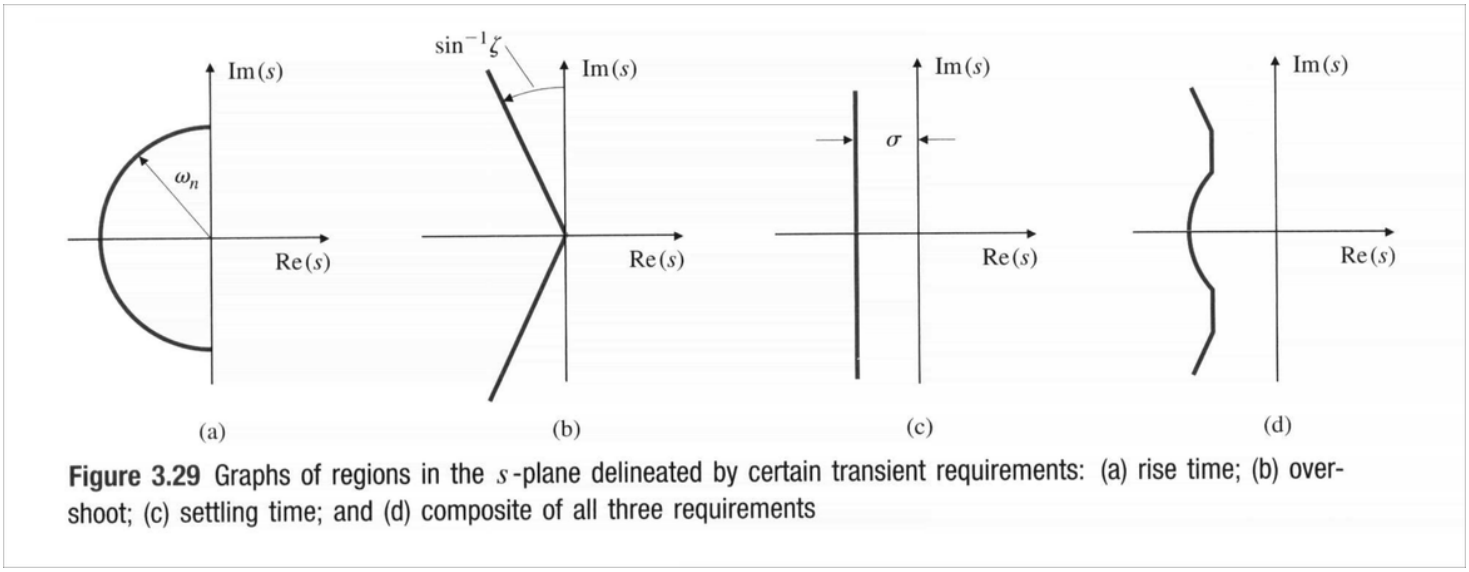


Figure 3.29 Graphs of regions in the s-plane delineated by certain transient requirements: (a) rise time; (b) overshoot; (c) settling time; and (d) composite of all three requirements

Figure 12: Regions on the S-plane created by transient specifications.

(a)	A maximum rise-time specification gives a minimum ω_n.
(b)	A maximum overshoot specification gives a minimum damping.
(c)	A maximum settling time specification gives a minimum α.
(d)	Putting all requirements together, gives the S-plane region in which the dominant-mode poles must lie.

4.1 Example

- Determine the region on the S-plane where poles may lie and meet these requirements:

  1. $t_s \leq 1.10$ [sec]
  2. $t_r \leq 0.35$ [sec]
  3. $M_p \leq 7\%$

1. The settling time requirement determines the minimum allowed α:

$$\alpha \geq \frac{4}{t_s} = 3.64 \quad [\text{sec}^{-1}] \tag{30}$$

2. The rise time requirement determines the minimum allowed ω_n:

$$\omega_n \geq \frac{1.8}{t_r} = 5.14 \ [\text{rad/sec}] \tag{31}$$

3. The overshoot requirement determines the minimum damping allowed, ρ:

$$\rho \geq \text{value given from plot} = 0.65 \tag{32}$$

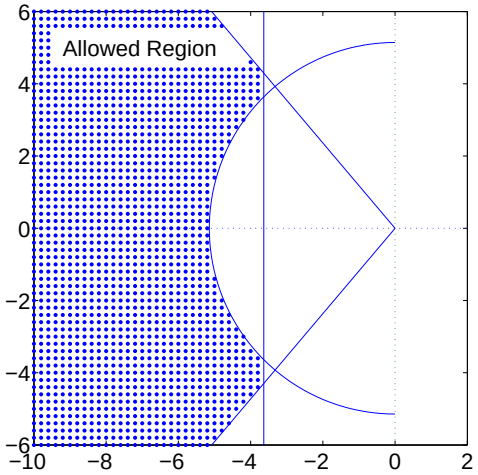


Figure 13: S-plane regions for dominant pole pair to satisfy requirements 1, 2 and 3.

5 The Effects of Additional Zeros and Poles on Response (Franklin *et al.* section 3.5 (7th ed.))

5.1 The Effects of Additional Zeros and Poles

- In the previous lectures we have developed tools to quantitatively connect pole-zero constellation to time-domain response characteristics.
- To get useful results, we considered a system with 2 poles and no zeros. The question naturally arises:
  - What happens if there are more poles and zeros ?
- The algebra in these cases (most real cases) is too complex to give any useful results
- Dimensional analysis can be used to investigate the effects of additional poles and zeros.
- In the end, we do not have exact design equations for most control systems (because the algebra is too complex), but use the dominant-mode approximation to get initial results, and iterative tuning to achieve final performance goals.

5.1.1 Dimensional Analysis

Enrichment Topic, Not Essential

- In dimensional analysis, we can simplify a nonlinear problem by choosing the primary units (time, length, mass, voltage, etc.) to make parameters of the problem come out be to 1.0.
  - Scale one parameter to 1.0 for each primary unit that can be rescaled.
  - In a problem with time, length, and mass we can scale out 3 parameters.
- Quantities can be measured in various units, for example:
  - The inertia of a vehicle might be 1,500 if measured in units of [Kg]
  - The inertia of the same vehicle is 1.0, if mass is measured in units of 1,500 [Kg].
- For this investigation, we will make the transfer function dimensionless by defining dimensionless time:

$$t^* = (t\,\omega_n) \;\; [\text{sec}]\,[\text{rad/sec}] = [\text{Dim}'\text{less}] \tag{33}$$

- Then every quantity with units of time will be rescaled, for example:
  - Variable s has units of $[\text{sec}^{-1}]$, so $s^* = (s/\omega_n) \;\; \frac{[[\text{sec}^{-1}]]}{[\text{rad/sec}]} = [\text{Dim}'\text{less}]$.
  - Variables α, ω and ω_n all have units of $[\text{sec}^{-1}]$ (or $[\text{rad/sec}]$), so:

$$\begin{aligned} \alpha^* &= \alpha/\omega_n = \rho \\ \omega^* &= \omega/\omega_n = \sqrt{1-\rho^2} \\ \omega_n^* &= \omega_n\,/\,\omega_n = 1.0 \end{aligned}$$

  - The natural variables are given from the dimensionless variables by:

$$\begin{aligned} t &= t^*\,/\,\omega_n \\ \alpha &= \alpha^*\,\omega_n \\ \omega &= \omega^*\,\omega_n \end{aligned}$$

- Now s^* and t^* are scaled to the speed of the system, with: $\omega_n^* = 1$.

- The dimensionless model is formed by forming "dimensionless groups" each place that a dimensioned term appears in the modeling equations.

- In this case groups of (s/ω_n), or s^*, are formed:

$$H(s) = \frac{b_1}{s^2 + 2\rho\,\omega_n\, s + \omega_n^2} = K_{rlg}\,\frac{1}{s^2 + 2\rho\,\omega_n\, s + \omega_n^2} \tag{34}$$

$$= \frac{K_{rlg}}{\omega_n^2}\,\frac{1}{(s/\omega_n)^2 + 2\rho\,(s/\omega_n) + 1} = \frac{K_{rlg}}{\omega_n^2}\,\frac{1}{s^{*2} + 2\rho\, s^* + 1} \tag{35}$$

- Now s^* and t^* are scaled to the speed of the system, with: $\omega_n^* = 1$.

5.1.1.1 Summary: Dimensional Analysis

- Dimensional analysis is a powerful tool to simplify nonlinear problems
  - The system of Eqn (34) is linear, but t_r, t_p, etc. relate to system parameters in a nonlinear way
  - You can remove one Degree of Freedom (DOF) from the problem for every dimension that can be rescaled
    - In this case the radius of the pole location is removed as a parameter (set to 1.0) by rescaling time
  - Dimensional analysis is used extensively in fluid mechanics, and other areas of non-linear dynamics

5.1.2 Effects of an additional zero

- In dim'less coordinates, the effect of adding a zero is seen in fig's 14 and 15.

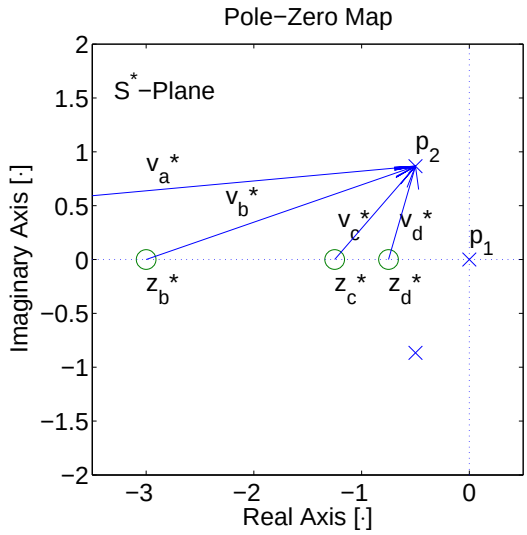


Figure 14: Pole-zero map on the dimensionless S-plane with $\rho = 0.5$. Zero at: z_a^* or z_b^* or z_c^* or z_d^*. Vectors from zeros to p_2 are: $\vec{v}_a^*$, $\vec{v}_b^*$, $\vec{v}_c^*$ and $\vec{v}_d^*$.

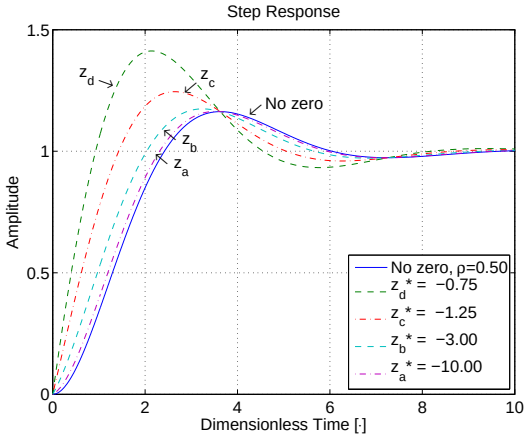


Figure 15: Step response of systems with no zero, and with one of: z_a^*, z_b^*, z_c^*or z_d^*.

1. A zero much faster than the Dom. mode (far to the left) has negligible effect.
2. The effect of the zero grows greater as it approaches the speed of the poles.
   - Rise time: grows faster;
   - Overshoot: increases;
   - Settling time: no influence.

Effects of an additional zero (continued)

- The first point above is very important:

  If additional poles and/or zeros are much faster than the dominant mode, they have negligible effect.

  – All systems have unmodeled high-speed dynamics (e.g., bending modes, speed of light delay).

  – But these dynamics play a negligible role if they are much faster than the dominant mode.

5.1.3 An additional zero, considering the graphical residue

- The negligible influence of fast poles and/or zeros can be understood by thinking of the residue and the graphical residue method.

- Putting in a zero at $z_a = -(b_2/b_1)$:

$$H(s) = \frac{b_1 s + b_2}{s^2 + 2\rho\omega_n s + \omega_n^2} = \frac{b_2}{\omega_n^2} \frac{\frac{s}{-z_a} + 1}{(1/\omega_n^2)(s^2 + 2\rho\omega_n s + \omega_n^2)} \quad (36)$$

  where $\left(b_2/\omega_n^2\right)$ is pulled out because $K_{DC} = b_2/\omega_n^2$.

- Forming the dimensionless groups:

$$H(s) = K_{DC} \frac{\frac{s/\omega_n}{-z_a/\omega_n} + 1}{(s/\omega_n)^2 + 2\rho\,(s/\omega_n) + 1} = K_{DC} \frac{s^*/(-z_a^*) + 1}{s^{*2} + 2\rho s^* + 1} \quad (37)$$

  where $z_a^* = z_a/\omega_n$ is the dimensionless zero location.

- Re-arrange to bring out the root-locus gain:

$$H(s) = \frac{K_{DC}}{-z_a^*} \frac{s^* - z_a^*}{s^{*2} + 2\rho s^* + 1} = K_{rlg} \frac{s^* - z_a^*}{s^{*2} + 2\rho s^* + 1} \quad (38)$$

- Eqn (38) shows that $K_{rlg} = (K_{DC}/-z_a^*)$

  $\longrightarrow$ **If K_{DC} is constant, then K_{rlg} gets smaller as $-z_a^*$ gets larger.**

Considering the graphical residue (continued)

- Now think of the graphical residue of the dominant mode:

$$C_2 = K_{rlg} \frac{\Pi \text{ vectors from zeros}}{\Pi \text{ vectors from poles}} \tag{39}$$

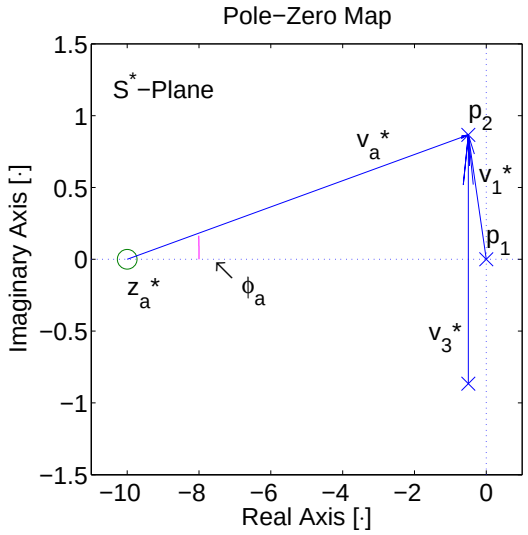


Figure 16: Pole-zero map showing z_a^*.

- The dominant mode of the response is given by:

$$C_2 = K_{rlg} \frac{\vec{v_a}}{\vec{v_1}\,\vec{v_3}} = \frac{K_{DC}}{-z_a^*} \frac{\left|\left|\vec{v_a^*}\right|\right| \angle \phi_a}{1 \angle \left(90^o + \sin^{-1}(\rho)\right) \quad 2\sqrt{1-\rho^2} \angle 90^o} \tag{40}$$

where $\left|\left|\vec{v_a^*}\right|\right| \angle \phi_a$ is vector $\vec{v_a^*}$ in polar coordinates.

- Which can be rewritten:

$$\begin{aligned} C_2 &= K_{DC} \frac{\left|\left|\vec{v_a^*}\right|\right|}{-z_a^*} \frac{1 \angle \phi_a}{2\sqrt{1-\rho^2} \angle \left(180^o + \sin^{-1}(\rho)\right)} \\ &= K_{DC} \frac{\left|\left|\vec{v_a^*}\right|\right|}{-z_a^*} \frac{1}{2\sqrt{1-\rho^2}} \angle \left(-180 - \sin^{-1}(\rho) + \phi_a\right) \end{aligned} \tag{41}$$

Influence of zero

- When $-z_a^* >> \alpha^*$, then $\left|\left|\vec{v_a^*}\right|\right| \approx -z_a^*$, and

$$C_2 \approx K_{DC} \frac{1}{2\sqrt{1-\rho^2}} \angle \left(-180 - \sin^{-1}(\rho) + \phi_a\right) \tag{42}$$

- When the zero is fast, K_{rlg} getting smaller approx. cancels $\left|\left|\vec{v_a^*}\right|\right|$ getting larger.
  - The influence of the zero is mainly through ϕ_a, the angle of $\vec{v_a}$.
  - If z_a is much faster than the dominant mode, angle ϕ_a will be small.
- Notice in figure 17, the initial slope of $y(t)$ increases as ϕ_a increases;
  - This corresponds to the angle of C_2 changing the vector as the zero changes position.
  - This reduces **rise time** and increases **overshoot**.

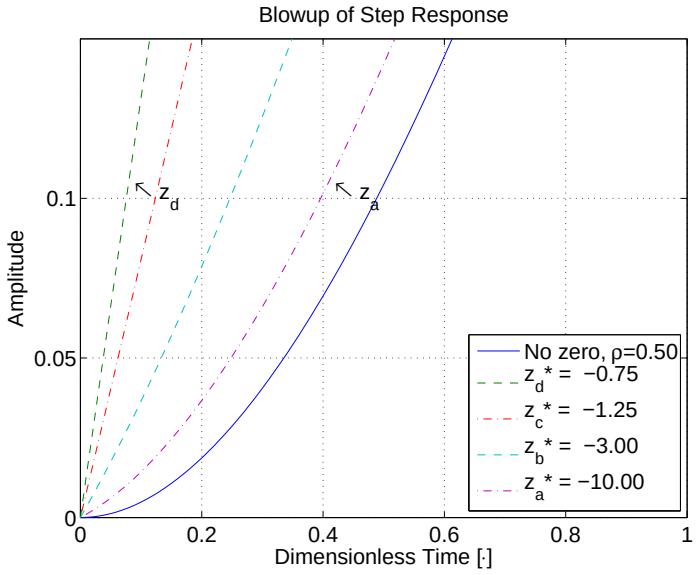


Figure 17: Blowup of step responses.

5.1.4 Summary: Effects of an additional zero

- The effects of an additional zero are summarized in tables 6 and 7.
- Table 6 shows that as the zero gets closer to the speed of the dominant mode, the initial response is faster, but with more overshoot.
- Table 7 shows the effects qualitatively.

Dim'less zero location:	No Zero	$z^* = -10$	$z^* = -3$	$z^* = -1.25$	$z^* = -0.75$
Rise Time: t_r	1.64	1.62	1.51	1.10	0.73
Peak Time: t_p	3.62	3.51	3.24	2.63	2.13
Percent Overshoot: M_p	16.30	16.39	17.43	24.51	41.26
Settling Time: t_s	8.00	8.00	8.00	8.00	8.00

Table 6: Performance measures for the responses of figure 15.

Dim'less zero location:	Zero far to left	As Zero Moves toward Dom. mode
Rise Time: t_r	No Effect	Faster
Peak Time: t_p	No Effect	Earlier
Percent Overshoot: M_p	No Effect	Increases
Settling Time: t_s	No Effect	No Effect

Table 7: Summary of qualitative effects of additional zero.

- The qualitative effect is most important, it lets us known how to adjust controller zeros on the S-plane to adjust system response.

Table 7 is the take-home result.

5.2 Effects of an additional pole

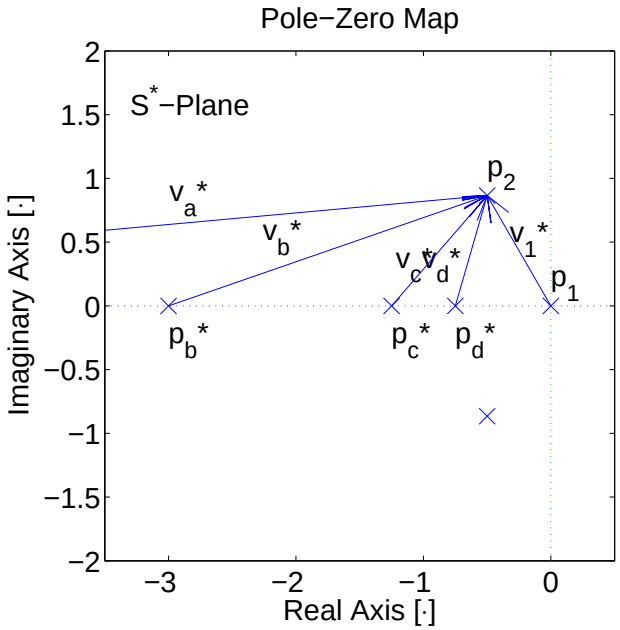


Figure 18: Pole-zero map on the dimensionless S-plane with $\rho = 0.5$. Extra pole at: p_a^* or p_b^* or p_c^* or p_d^*. Vectors from extra poles to p_2 are: $\vec{v}_a^*$, $\vec{v}_b^*$, $\vec{v}_c^*$ and $\vec{v}_d^*$.

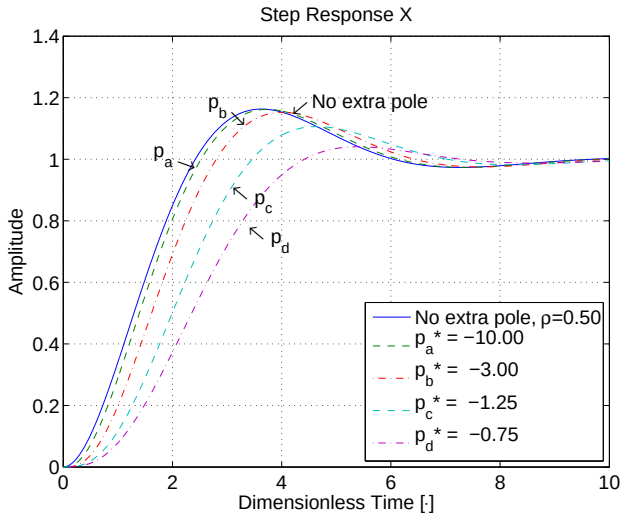


Figure 19: Step response of systems with no extra pole, and with one of: p_a^* or p_b^* or p_c^* or p_d^*.

5.3 Summary: Effects of a zero or an additional pole

- Examining figure 19, we can determine the qualitative effects of an additional pole, listed in table 8.

Dim'less zero location:	Additional Pole far to left	Additional Pole Moves toward Dominant Mode (in from left)	As Zero Moves toward Dominant mode (in from left)
Rise Time: t_r	No Effect	Slower	Faster
Peak Time: t_p	No Effect	Later	Earlier
Percent Overshoot: M_p	No Effect	Decreases	Increases
Settling Time: t_s	No Effect	No Effect	No Effect

Table 8: Summary of qualitative effects of an additional pole (with zero repeated).

- Table 8 is useful for design, because it shows qualitatively the impact of:
  - Adding a pole or zero to the system, or
  - Shifting a zero or a non-dominant pole to the left or right.

6 Stability and the Routh stability criterion

6.1 Stability

- There are a number of formal mathematical definitions of stability used for advanced control theory.

6.1.1 Global Asymptotic Stability

- The definition commonly applied to linear systems is ***Global Asymptotic Stability***, which requires that:

$$y(t) \to y_{ss} \; as \; t \to \infty \,, \quad \forall y(0) \tag{43}$$

- A linear time-invariant system is globally asymptotically stable (or simply 'stable') if all of the roots of the transfer function denominator polynomial have negative real parts (i.e., all poles are in the *strict* left-half plane).

- A linear time-invariant system is unstable otherwise.

6.1.2 Neutral stability

- A ***neutrally stable*** (or ***marginally stable***) system has one or more non-repeated poles on the $j\omega$ axis.

  – The output of such a system does not blowup, but does not go to zero either.

  – Repeated poles on the $j\omega$ axis (such as $1/s^2$) make a system unstable, because of the

$$y(t) = \cdots + c_1 t e^{-0t} + \cdots \tag{44}$$

  term that shows up in $y(t)$.

6.1.3 BIBO stability

- A system is ***Bounded-Input, Bounded-Output*** (***BIBO***) stable if a bounded input will always give a bounded output.
  - The output must be bounded, but is not required to converge to a steady-state value.
  - Think about friction, if one digs into the nitty-gritty, many mechanical systems are BIBO stable.

6.2 Routh stability criterion (Franklin *et al.* section 3.6 (7th ed.))

Enrichment Topic

- To determine stability it is necessary to find the roots of the characteristic equation polynomial.
- It is difficult to find the roots of a polynomial of greater than 2nd order by hand.
- In the days before digital computers, a very remarkable procedure existed to determine whether there were any roots of a polynomial in the right-half plane:
  - The Routh (or Routh-Hurwitz) stability criterion.
    - Discovered by Routh in 1874;
    - Independently discovered by Hurwitz in 1895.
- Today, testing the stability of high-order transfer functions by hand is not very common.

Routh stability criterion (continued)

- The Routh stability criterion is still sometimes used for analysis or theory, because it can produce symbolic results, for example:

  – 2nd order:

  $$H(s) = \frac{b_1 s + b_2}{s^2 + a_1 s + a_2}$$

  Routh stability criterion: stable if and only if:

  $a_1 > 0$ and $a_2 > 0$.

  – 3rd order:

  $$H(s) = \frac{b_1 s^2 + b_2 s + b_3}{s^3 + a_1 s^2 + a_2 s + a_3}$$

  Routh stability criterion: stable if and only if:

  $a_1, a_2, a_3 > 0$, and
  $a_3 < a_1 a_2$ (if $a_3 = a_1 a_2$, the system will be neutrally stable)

  – n^{th} order:

  $$H(s) = \frac{b_1 s^m + \cdots + b_m s + b_{m+1}}{a_0 s^n + a_1 s^{n-1} + \cdots + a_{n-1} s + a_n} \qquad (45)$$

  Full Routh-Hurwitz array needed to determine stability, but we can say from the Routh stability criterion: one condition in which the system will be unstable is:

  $$a_0 > 0 \text{ , and } \quad \text{any } a_i < 0 \text{ .}$$

7 Study examples

- Study examples are provided for consideration of the relations between pole-zero constellation and system response derived in this chapter.
- Consider and discuss the modes of each.

7.1 Example 1

$$T_{ry}(s) = \frac{7.91\,s + 316.2}{s^3 + 16.74\,s^2 + 107.7\,s + 316.2}$$

$$\begin{aligned} & & -8.415 \\ \text{Poles:} & & -4.163 + 4.499\,j \\ & & -4.163 - 4.499\,j \\ \text{Zeros:} & & -40.0 \end{aligned}$$

$$\alpha = 4.163\,, \qquad \omega = 4.499\,, \qquad \omega_n = 6.130$$

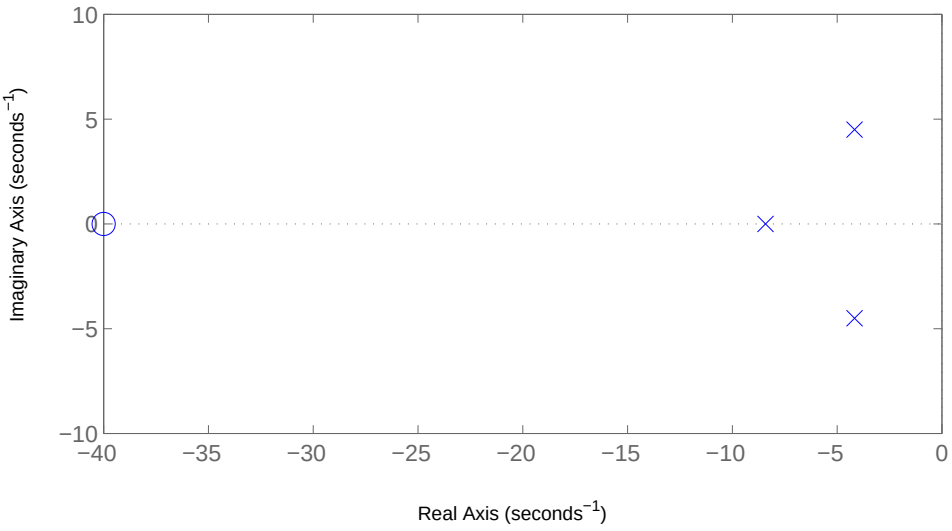


Figure 20: Pole-zero map.

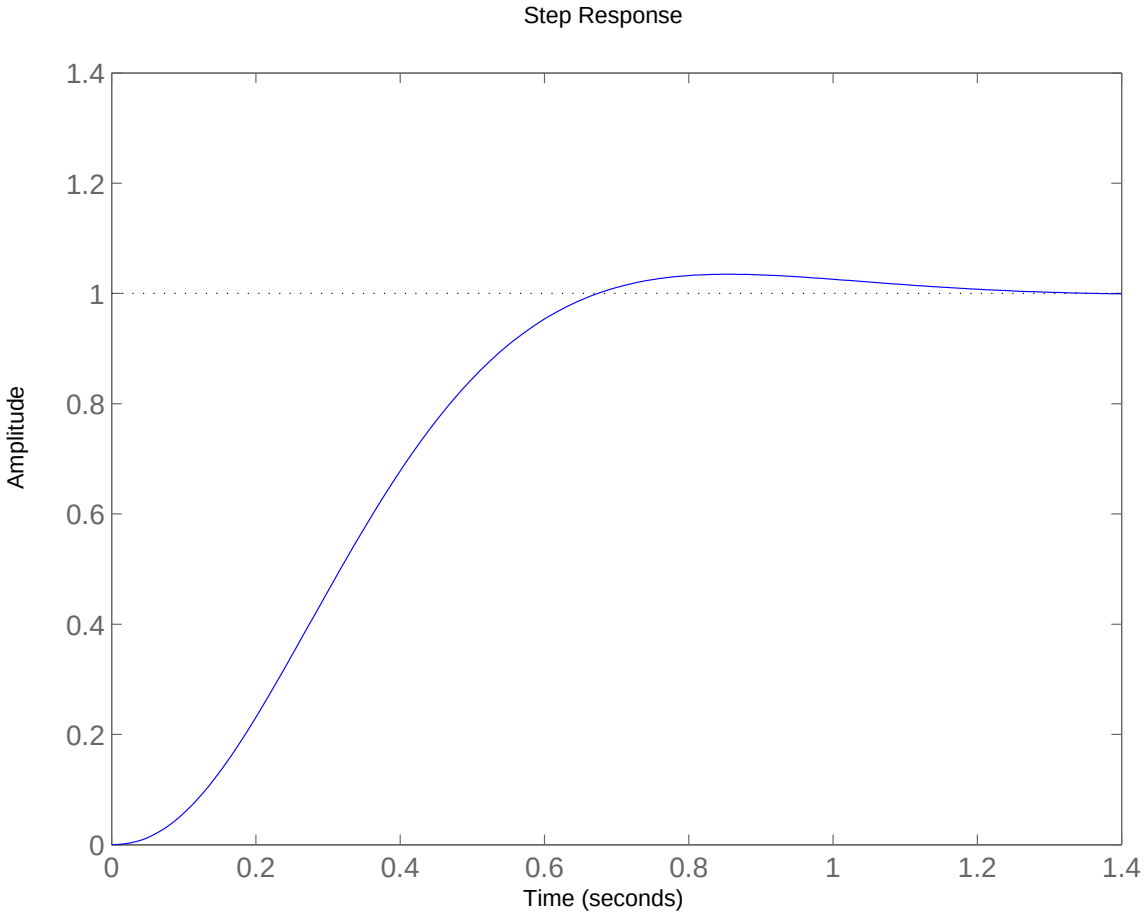


Figure 21: Step response.

- What is the dominant mode ?
- What extra pole or zero perceptibly effects the response ?
  - What are the effects of the extra pole or zero ?
- What extra pole or zero has no apparent effect on the response ?

7.2 Example 2

$$T_{ry}(s) = \frac{120\,s + 1000}{0.024\,s^3 + 2.18\,s^2 + 136.2\,s + 1100}$$

$$\begin{aligned} & & -9.3254 \\ \text{Poles:} & & -40.754 + 57.044\,j \\ & & -40.754 - 57.044\,j \\ \text{Zeros:} & & -8.333 \end{aligned}$$

$$\alpha = 40.754\,, \qquad \omega = 57.044\,, \qquad \omega_n = 70.106$$

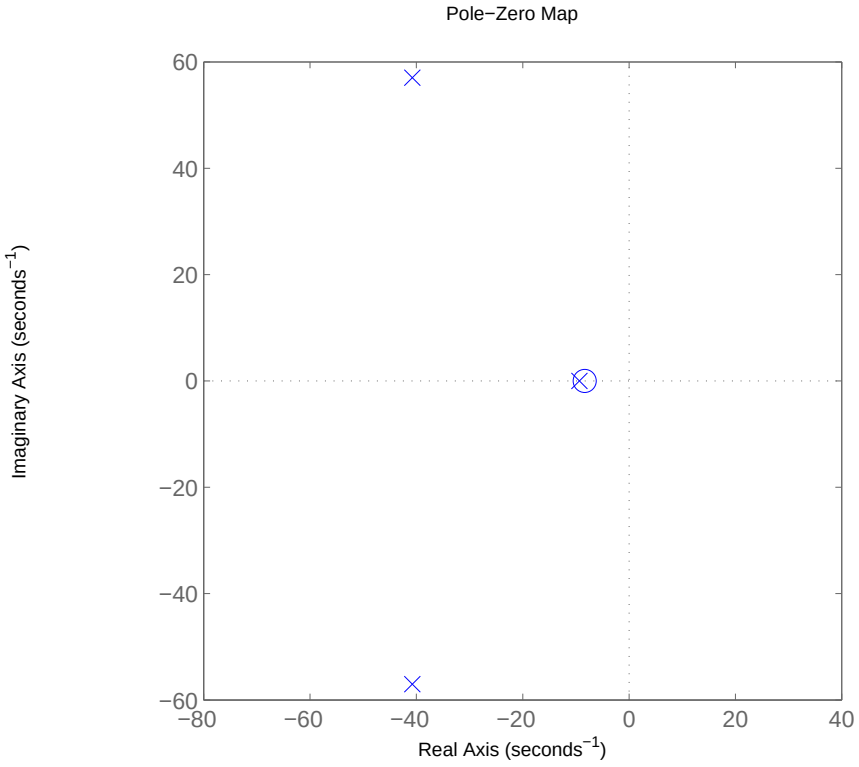


Figure 22: Pole-zero map.

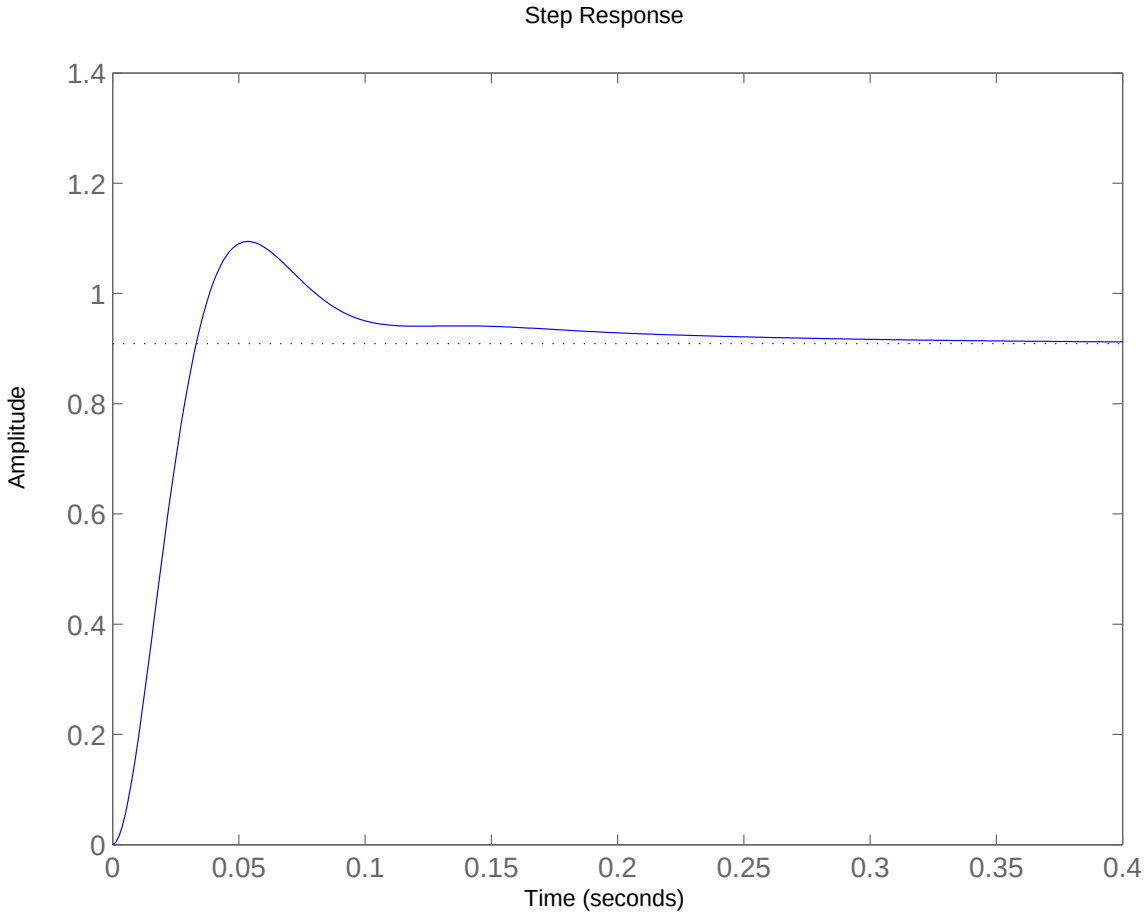


Figure 23: Step response.

- What is the dominant mode ?
- Is this the slowest mode ?
- If not, why not ?
- What aspect of the response is the result of the slow real mode ?

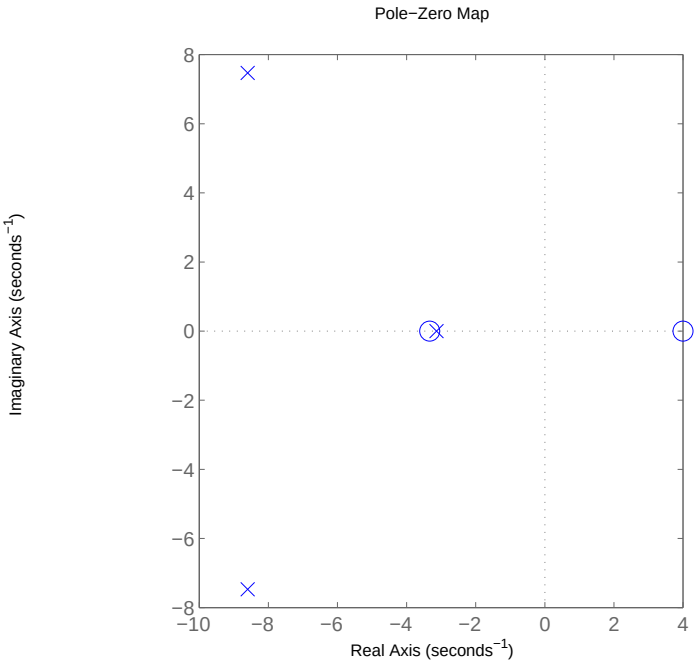
7.3 Example 3

$$T_{ry}(s) = \frac{-1.08\,s^2 + 0.72\,s + 14.4}{0.06\,s^3 + 1.22\,s^2 + 11.02\,s + 24.4}$$

$$\text{Poles:} \quad \begin{matrix} -3.1338 \\ -8.600 + 7.471\,j \\ -8.600 - 7.471\,j \end{matrix}$$

$$\text{Zeros:} \quad \begin{matrix} 4.000 \\ -3.333 \end{matrix}$$

$$\alpha = 8.600\,, \qquad \omega = 7.471\,, \qquad \omega_n = 11.391$$



Right-half plane zero
$\rightarrow$
non-minimum phase system.

Figure 24: Pole-zero map.

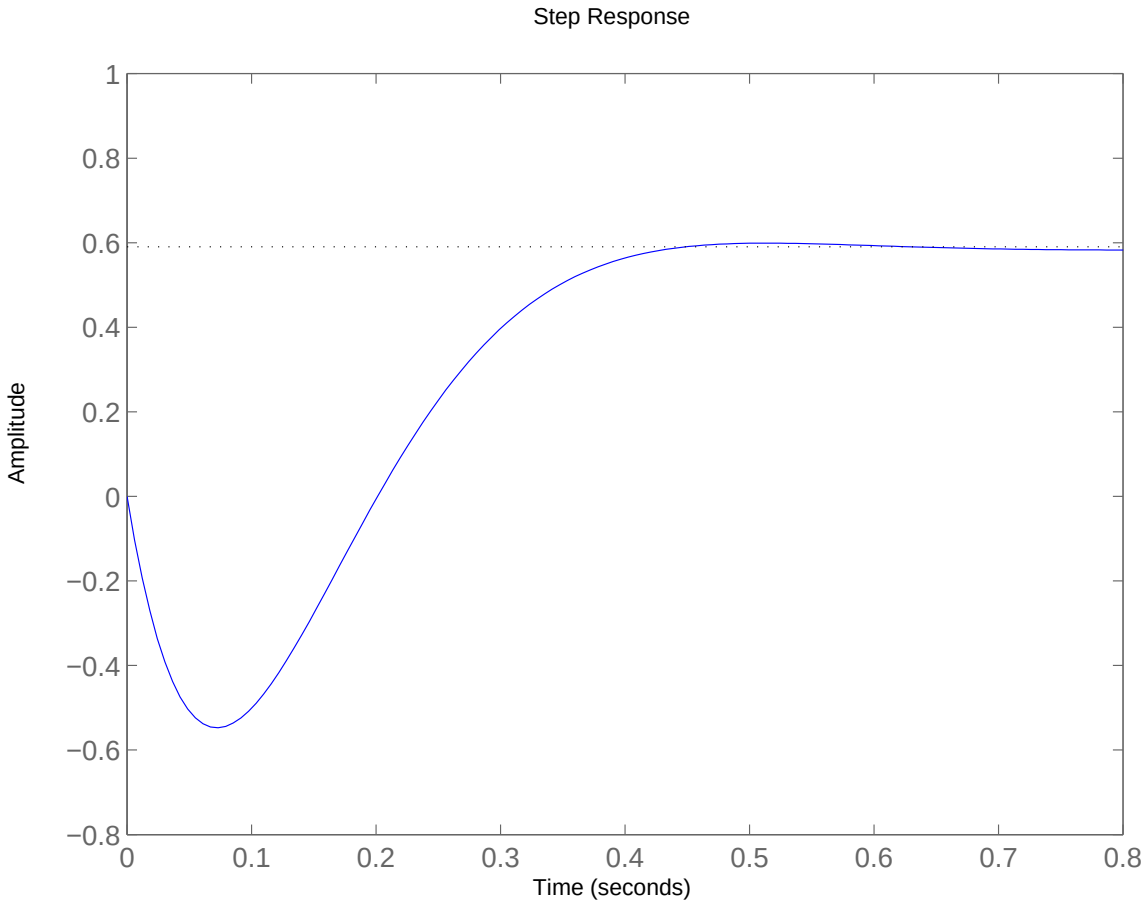


Figure 25: Step response, the negative-going transient is typical of non-minimum phase systems.

- What is the dominant mode ?
- What is the effect of the right-half-plane zero ?
- What is the name for this type of system ?

7.4 Example 4

$$T_{ry}(s) = \frac{320.2}{s^2 + 39.6s + 320.2}$$

$$\text{Poles:} \quad \begin{matrix} -11.091 \\ -28.866 \end{matrix}$$

Zeros: none

$$\alpha = 11.09$$

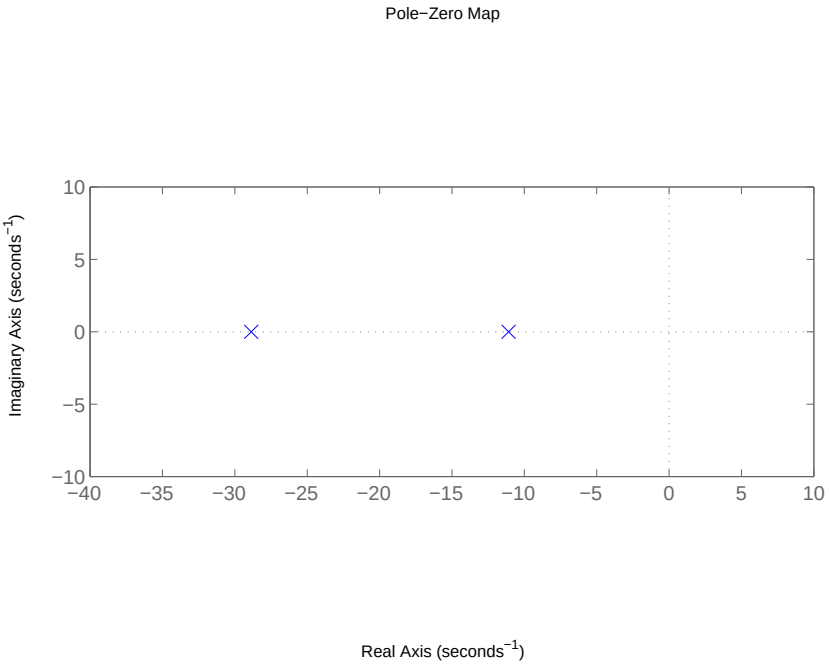


Figure 26: Pole-zero map.

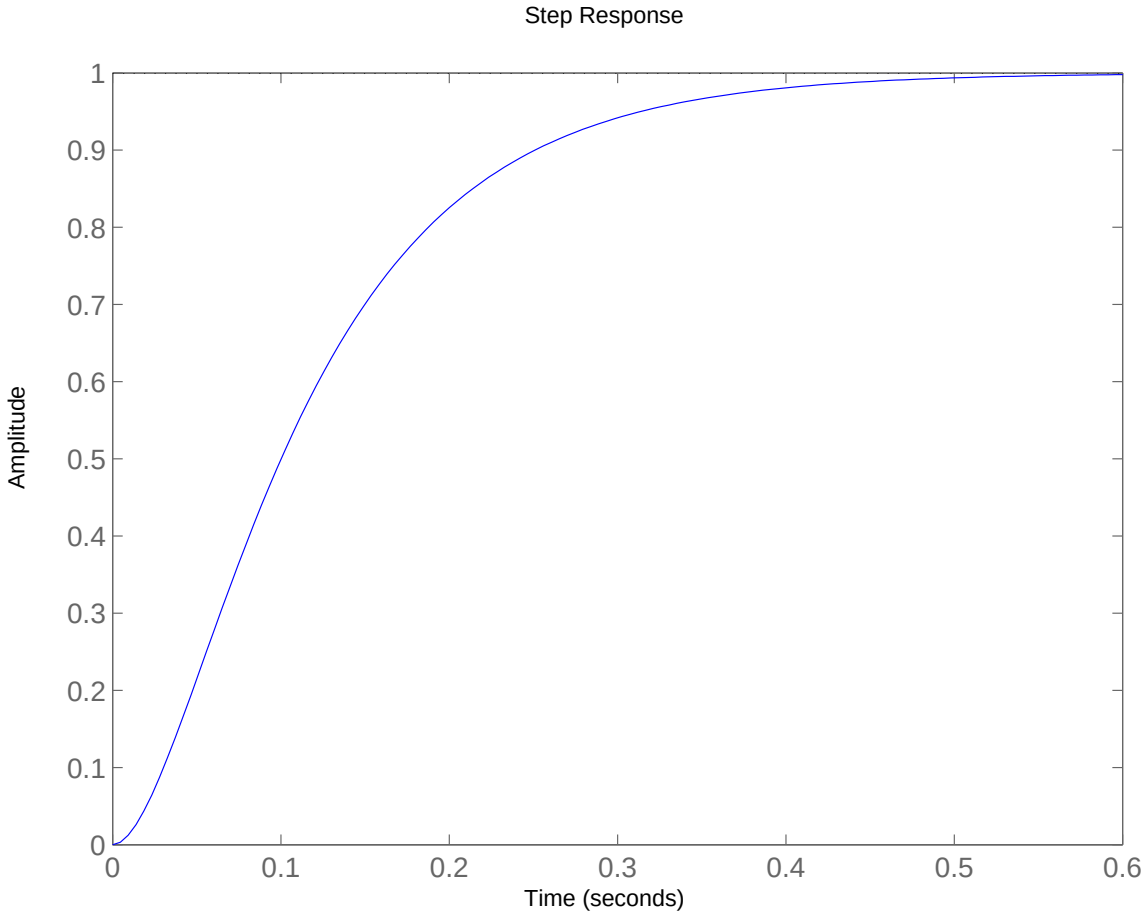


Figure 27: Step response.

- What is the dominant mode ?
- What first order system has nearly the same step response as this system ?

8 Conclusions

- Dimensional analysis is a powerful tool for simplifying non-linear problems.
- Doing controller design, we will be able to place controller zeros on the S-plane, and move around all of the controller+plant poles.
- Using
  - Qualitative knowledge of what happens to the system response when poles and zeros are adjusted

  Allows us to determine:
  - Which direction to adjust parameters to fine-tune the system response, for example, we might move a zero to be nearer the dominant poles.

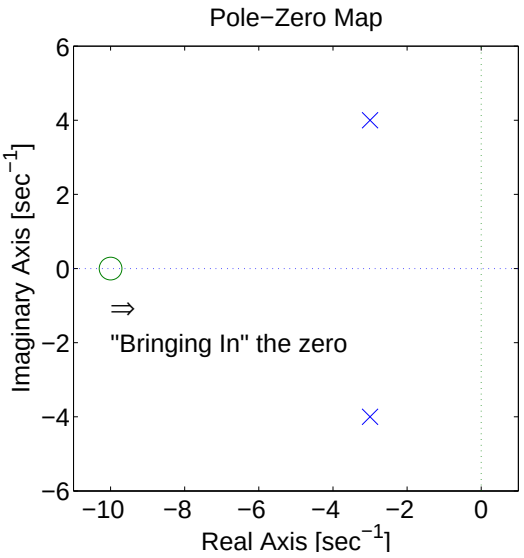


Figure 28: “Bringing In” the zero.

- The Routh-Array is not used day-to-day to determine stability, but can still be used to get symbolic or theoretical results.